

X1  
DigitalCarrierJack.SchDoc

Encoder1  
Encoder1.SchDoc

Encoder2  
Encoder2.SchDoc

Encoder3  
Encoder3.SchDoc

Connector  
Connectors.SchDoc

DIG\_IO  
PowerSupply  
DIG\_Res  
Status  
I2C  
CPLD\_IO

DIG\_IO  
PowerSupply  
DIG\_Res  
Status  
I2C  
CPLD\_IO

Encoder\_Signals\_1  
Encoder\_Signals\_2  
Encoder\_Signals\_3

Encoder\_Signals

Encoder\_Signals

Encoder\_Signals

GND  
GND



Revision

Serial  
Serial

UltraZohm\_DigitalVoltage  
Project

Designer

Date

|                         |                                                 |                 |          |                                                                                                   |      |                                                                                       |
|-------------------------|-------------------------------------------------|-----------------|----------|---------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------|
| Title                   |                                                 | TopSheet.SchDoc |          | Technische Hochschule Nürnberg Georg Simon Ohm<br>Fürther Straße 250<br>90429 Nürnberg<br>GERMANY |      |  |
| Size:                   | A4                                              | Revision:       | 1v00     |                                                                                                   |      |                                                                                       |
| Date:                   | 09.03.2022                                      | Time:           | 11:14:19 | Sheet 1                                                                                           | of 6 | TECHNISCHE HOCHSCHULE NÜRNBERG<br>Georg Simon Ohm                                     |
| Project:                | UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB |                 |          |                                                                                                   |      |                                                                                       |
| Author: Michaela Hlatky |                                                 |                 |          |                                                                                                   |      |                                                                                       |

A

B

C

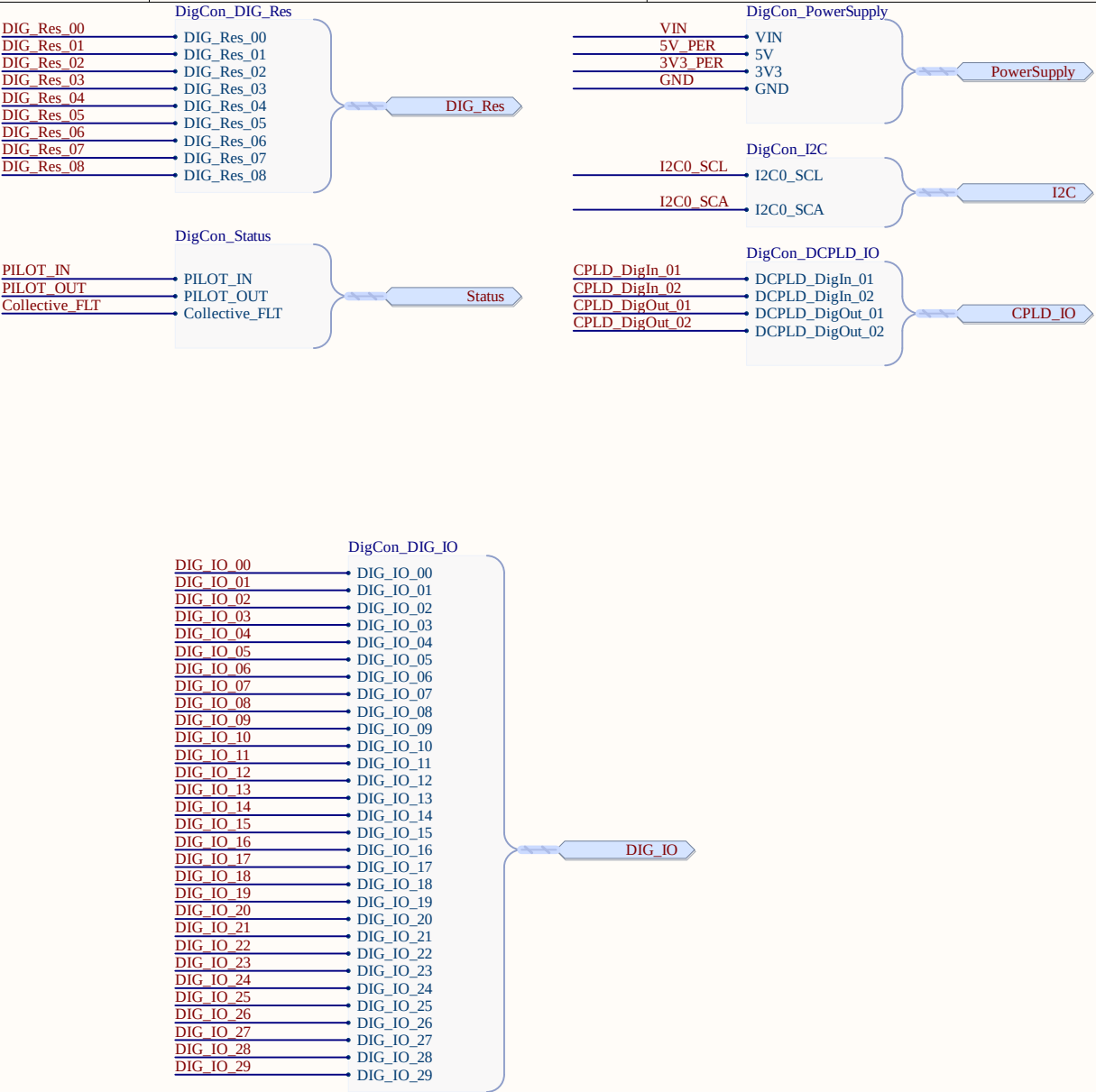
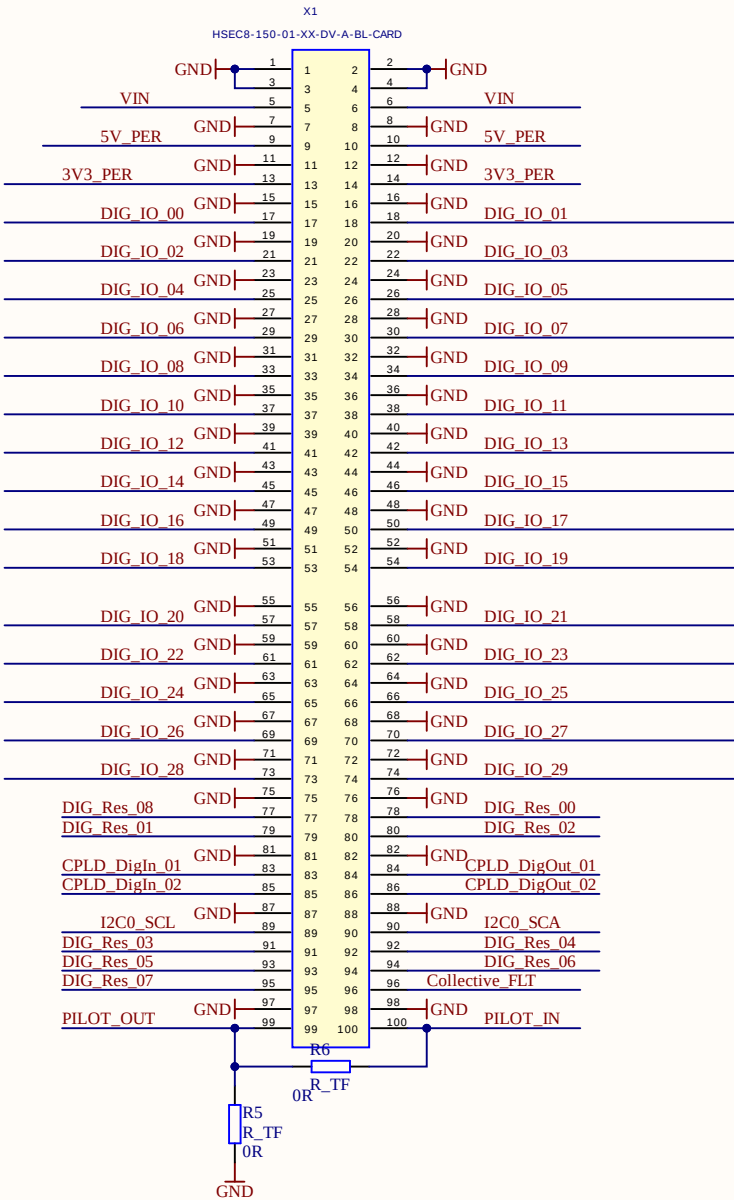
D

A

B

C

D



A

A

B

B

C

C

D

D

## DigCon\_DIG\_IO

|           |           |
|-----------|-----------|
| DIG_IO_00 | DIG_IO_00 |
| DIG_IO_01 | DIG_IO_01 |
| DIG_IO_02 | DIG_IO_02 |
| DIG_IO_03 | DIG_IO_03 |
| DIG_IO_04 | DIG_IO_04 |
| DIG_IO_05 | DIG_IO_05 |
| DIG_IO_06 | DIG_IO_06 |
| DIG_IO_07 | DIG_IO_07 |
| DIG_IO_08 | DIG_IO_08 |
| DIG_IO_09 | DIG_IO_09 |
| DIG_IO_10 | DIG_IO_10 |
| DIG_IO_11 | DIG_IO_11 |
| DIG_IO_12 | DIG_IO_12 |
| DIG_IO_13 | DIG_IO_13 |
| DIG_IO_14 | DIG_IO_14 |
| DIG_IO_15 | DIG_IO_15 |
| DIG_IO_16 | DIG_IO_16 |
| DIG_IO_17 | DIG_IO_17 |
| DIG_IO_18 | DIG_IO_18 |
| DIG_IO_19 | DIG_IO_19 |
| DIG_IO_20 | DIG_IO_20 |
| DIG_IO_21 | DIG_IO_21 |
| DIG_IO_22 | DIG_IO_22 |
| DIG_IO_23 | DIG_IO_23 |
| DIG_IO_24 | DIG_IO_24 |
| DIG_IO_25 | DIG_IO_25 |
| DIG_IO_26 | DIG_IO_26 |
| DIG_IO_27 | DIG_IO_27 |
| DIG_IO_28 | DIG_IO_28 |
| DIG_IO_29 | DIG_IO_29 |

DIG\_IO

## Encoder\_Signals\_CH3

|           |            |
|-----------|------------|
| A_encoder | A_encoder3 |
| B_encoder | B_encoder3 |
| Z_encoder | Z_encoder3 |

Encoder\_Signals\_3

## Encoder\_Signals\_CH2

|           |            |
|-----------|------------|
| A_encoder | A_encoder2 |
| B_encoder | B_encoder2 |
| Z_encoder | Z_encoder2 |

Encoder\_Signals\_2

## Encoder\_Signals\_CH1

|           |            |
|-----------|------------|
| A_encoder | A_encoder1 |
| B_encoder | B_encoder1 |
| Z_encoder | Z_encoder1 |

Encoder\_Signals\_1

|           |    |    |              |
|-----------|----|----|--------------|
| DIG_IO_00 | 01 | 02 | 02 DIG_IO_02 |
| DIG_IO_01 | 04 | 05 | 05 DIG_IO_04 |
| DIG_IO_03 | 07 | 08 | 08 DIG_IO_06 |
| DIG_IO_05 | 10 | 11 | 11 DIG_IO_08 |
| DIG_IO_07 | 13 | 14 | 14 DIG_IO_10 |
| DIG_IO_09 | 16 | 17 | 17 DIG_IO_12 |
| DIG_IO_11 | 19 | 20 | 20 DIG_IO_14 |
| DIG_IO_13 | 22 | 23 | 23 DIG_IO_16 |
| DIG_IO_15 | 25 | 26 | 26 DIG_IO_18 |
| DIG_IO_17 | 28 | 29 | 29 DIG_IO_19 |

GND

X2

03

06

09

12

15

18

21

24

27

30

Pin Header

X4

|               |    |    |                  |
|---------------|----|----|------------------|
| VIN_fused     | 01 | 02 | 02 VIN_fused     |
| GND           | 03 | 04 | GND              |
| 5V_PER_fused  | 05 | 06 | 06 5V_PER_fused  |
| GND           | 07 | 08 | GND              |
| 3V3_PER_fused | 09 | 10 | 10 3V3_PER_fused |
| GND           | 11 | 12 | GND              |

Pin Header

## DigCon\_DIG\_Res

|            |            |
|------------|------------|
| DIG_Res_00 | DIG_Res_00 |
| DIG_Res_01 | DIG_Res_01 |
| DIG_Res_02 | DIG_Res_02 |
| DIG_Res_03 | DIG_Res_03 |
| DIG_Res_04 | DIG_Res_04 |
| DIG_Res_05 | DIG_Res_05 |
| DIG_Res_06 | DIG_Res_06 |
| DIG_Res_07 | DIG_Res_07 |
| DIG_Res_08 | DIG_Res_08 |

DIG\_Res

## DigCon\_DCPLD\_IO

|                 |                |
|-----------------|----------------|
| DCPLD_DigIn_01  | CPLD_DigIn_01  |
| DCPLD_DigIn_02  | CPLD_DigIn_02  |
| DCPLD_DigOut_01 | CPLD_DigOut_01 |
| DCPLD_DigOut_02 | CPLD_DigOut_02 |

CPLD\_IO

## DigCon\_Status

|                |                |
|----------------|----------------|
| PILOT_IN       | PILOT_IN       |
| PILOT_OUT      | PILOT_OUT      |
| Collective_FLT | Collective_FLT |

Status

|               |    |    |                   |
|---------------|----|----|-------------------|
| DIG_IO_21     | 01 | 02 | 02 DIG_IO_20      |
| DIG_IO_23     | 04 | 05 | 05 DIG_IO_22      |
| DIG_IO_25     | 07 | 08 | 08 DIG_IO_24      |
| DIG_IO_27     | 10 | 11 | 11 DIG_IO_26      |
| DIG_IO_29     | 13 | 14 | 14 DIG_IO_28      |
| DIG_Res_07    | 16 | 17 | 17 DIG_Res_08     |
| CPLD_DigIn_01 | 19 | 20 | 20 CPLD_DigOut_01 |
| CPLD_DigIn_02 | 22 | 23 | 23 CPLD_DigOut_02 |
| I2C0_SCL      | 25 | 26 | 26 I2C0_SCA       |
| PILOT_OUT     | 28 | 29 | 29 PILOT_IN       |

GND

X3

03

06

09

12

15

18

21

24

27

30

Pin Header

F1

|                     |               |
|---------------------|---------------|
| VIN                 | VIN_fused     |
| Littelfuse_PTC_1812 |               |
| F2                  |               |
| 5V_PER              | 5V_PER_fused  |
| Littelfuse_PTC_1812 |               |
| F3                  |               |
| 3V3_PER             | 3V3_PER_fused |
| Littelfuse_PTC_1812 |               |

## DigCon\_I2C

|          |          |
|----------|----------|
| I2C0_SCL | I2C0_SCL |
| I2C0_SCA | I2C0_SCA |

I2C

## DigCon\_PowerSupply

|     |         |
|-----|---------|
| VIN | VIN     |
| 5V  | 5V_PER  |
| 3V3 | 3V3_PER |
| GND | GND     |

PowerSupply

A

B

C

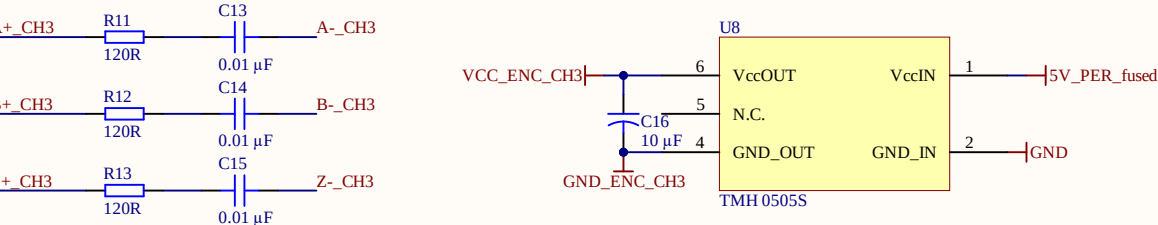
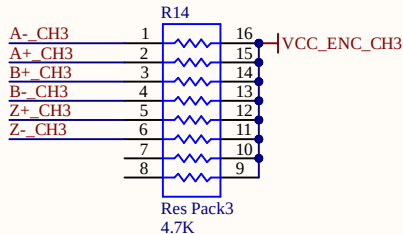
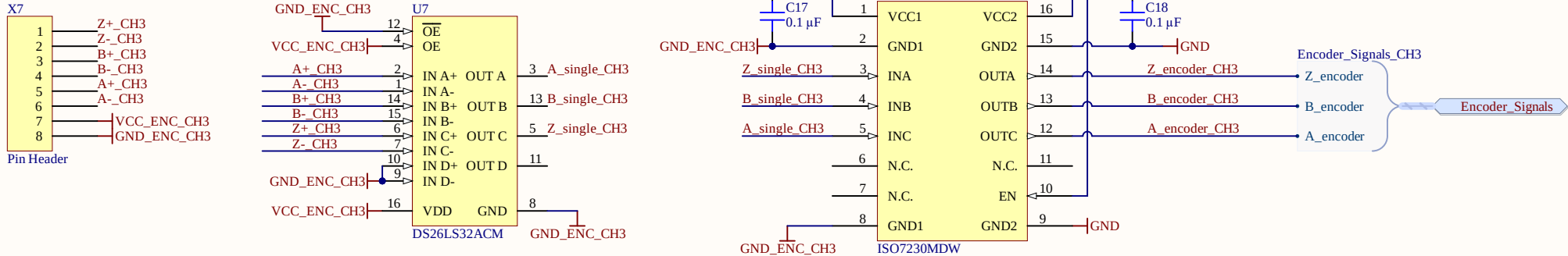
D

A

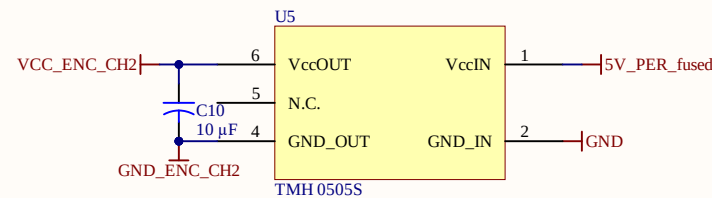
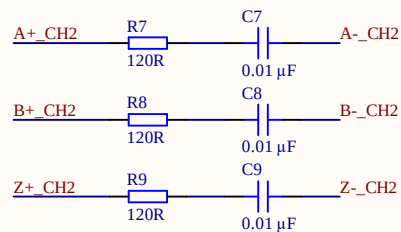
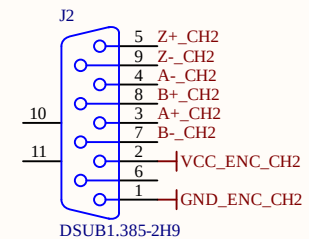
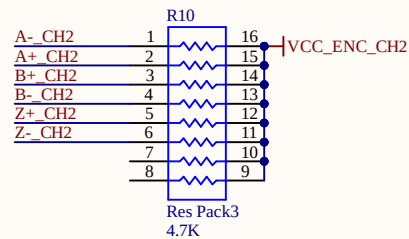
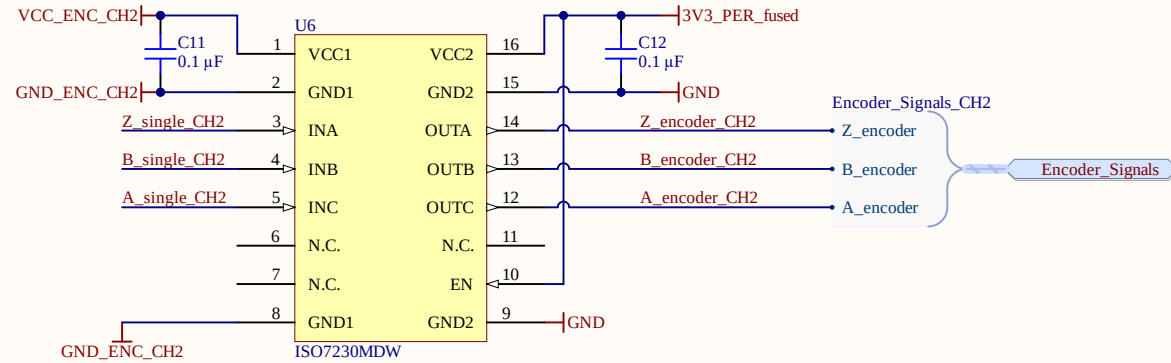
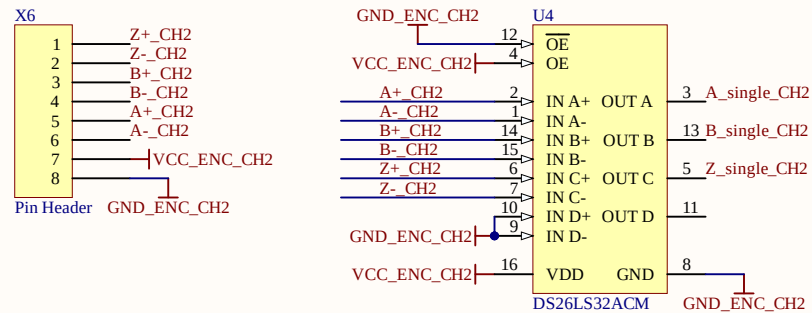
B

C

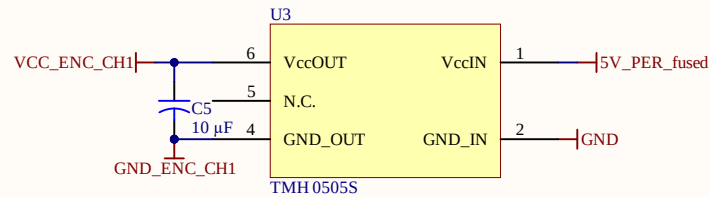
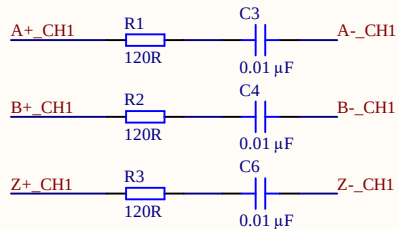
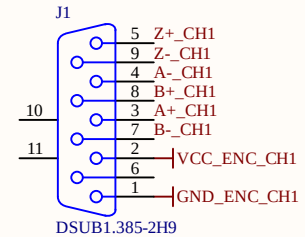
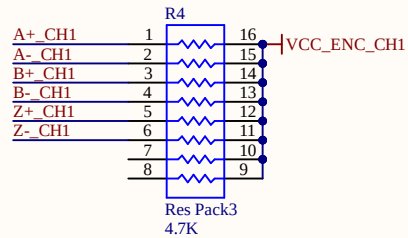
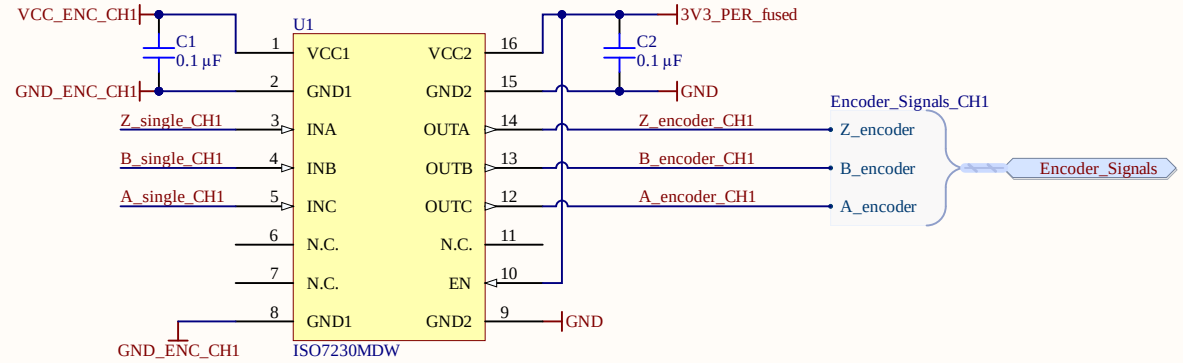
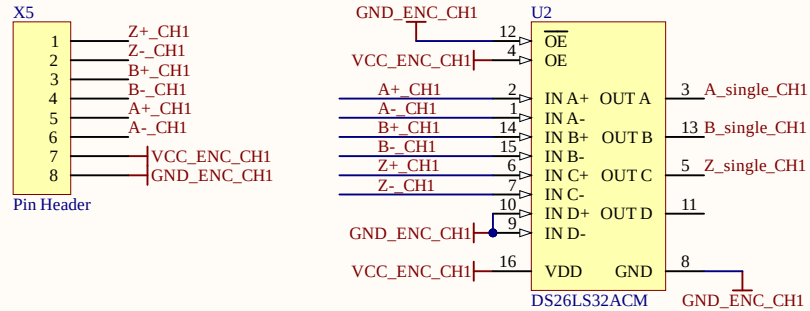
D



| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| U0+     | Pin3 | 0+      | blue          |
| U0-     | Pin4 | 0-      | red           |
| U1+     | Pin8 | A+      | green         |
| U1-     | Pin7 | A-      | yellow        |
| U2+     | Pin5 | B+      | grey          |
| U2-     | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |



| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| U0+     | Pin3 | 0+      | blue          |
| U0-     | Pin4 | 0-      | red           |
| U1+     | Pin8 | A+      | green         |
| U1-     | Pin7 | A-      | yellow        |
| U2+     | Pin5 | B+      | grey          |
| U2-     | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |



| Label   | Pin  | Encoder | Color(Kübler) |
|---------|------|---------|---------------|
| U0+     | Pin3 | 0+      | blue          |
| U0-     | Pin4 | 0-      | red           |
| U1+     | Pin8 | A+      | green         |
| U1-     | Pin7 | A-      | yellow        |
| U2+     | Pin5 | B+      | grey          |
| U2-     | Pin9 | B-      | pink          |
| VCC_ENC | Pin2 | UB      | brown         |
| GND_ENC | Pin1 | 0V      | white         |
| NC      | Pin6 | --      |               |